

Toward Intelligent Cauliflower Disease Recognition: A CLAHE-Enhanced Image Processing and SVM-Based Classification Framework

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Abstract—Accurate and timely identification of cauliflower diseases is essential for reducing crop losses and supporting effective disease management. However, conventional visual diagnosis is labor-intensive, time-consuming, and susceptible to human error. This study proposes an automated image-based framework for multi-class cauliflower disease recognition by integrating image enhancement, segmentation, handcrafted feature extraction, and machine learning. The proposed framework first applies CLAHE to enhance local image contrast and improve the visibility of disease-related patterns. K-means clustering is subsequently employed to segment relevant image regions, followed by the extraction of texture and statistical features using the Gray-Level Co-occurrence Matrix. The resulting feature representations are classified using a SVM to distinguish among five classes of cauliflower disease. Experiments are conducted on a publicly available dataset containing 656 RGB images collected under varying illumination, leaf orientations, and background conditions. The dataset was divided into training, validation, and testing subsets using a 70:10:20 ratio, and model reliability is further assessed using five-fold stratified cross-validation. The proposed framework achieved an overall accuracy of 97.35%, with precision, recall, and F1-score of 96.35%, 97.35%, and 97.35%, respectively. Comparative evaluation against existing cauliflower disease recognition methods demonstrated the competitive performance of the proposed approach. These findings indicate that the integration of CLAHE, segmentation, GLCM-based feature extraction, and SVM provides an effective and computationally practical approach for automated cauliflower disease recognition and can serve as a decision-support tool for agricultural disease management.

Index Terms—Cauliflower Disease, Support Vector Machine, Machine Learning, Image Processing, CLAHE.

I. INTRODUCTION

Agriculture is essential for food security and economic development, especially in developing countries. Cauliflower is a widely grown vegetable crop due to its nutritional and commercial value, but its production is often threatened by diseases. These diseases negatively impact yield and quality, leading to significant economic losses. Therefore, early and accurate detection is crucial for effective disease management and sustainable cauliflower production [1]. Typically, disease diagnosis is made by the visual sighting of agricultural professionals. It is a manual, labour-intensive and time-consuming procedure, particularly on larger farms. Automated plant disease recognition systems have become possible due to the recent developments in Computer Vision (CV) and Machine Learning (ML). There are a few studies that have used ML

and Deep Learning (DL) to classify diseases in agricultural crops.

Although the above improvements have been made, current methods present certain drawbacks, including inadequate dataset size, inadequate feature extraction, poor segmentation, and low classification quality. Furthermore, most literature deals with a limited set of categories of diseases and uses DL models that are not suitable for practical application in the agricultural environment of resource-limited areas such as rural regions of developing countries [2], [3]. Other kinds of Cauliflower are shown in Fig. 1.

To address these challenges, this study introduces an automated framework for cauliflower disease recognition by integrating image processing techniques with ML. The proposed pipeline begins with image preprocessing to improve visual quality, followed by segmentation to isolate the affected regions. Texture characteristics are then extracted using the Gray-Level Co-occurrence Matrix (GLCM), together with statistical descriptors that effectively represent disease-specific patterns. Finally, a Support Vector Machine (SVM) classifier is employed to distinguish among different cauliflower disease categories and healthy samples. The primary contributions of this work are summarized as follows:

- Develops an integrated and resource-efficient framework for multi-class cauliflower disease recognition by combining contrast enhancement, disease-region segmentation, feature extraction, and SVM-based classification for automated identification of multiple disease categories.
- Performs a systematic evaluation of the individual and combined effects of the preprocessing and feature-extraction components through controlled experiments, including CLAHE, and GLCM-based features, to identify their contribution to classification performance.
- Provides a comprehensive and reproducible evaluation of the proposed framework using stratified data partitioning, class-wise performance metrics, and confusion-matrix analysis, while comparing the proposed approach with conventional ML baselines to assess its effectiveness and computational practicality.

II. RELATED WORK

Automated plant disease detection has become an important research area in recent years, primarily because of its ability to enhance agricultural productivity and minimize crop damage.

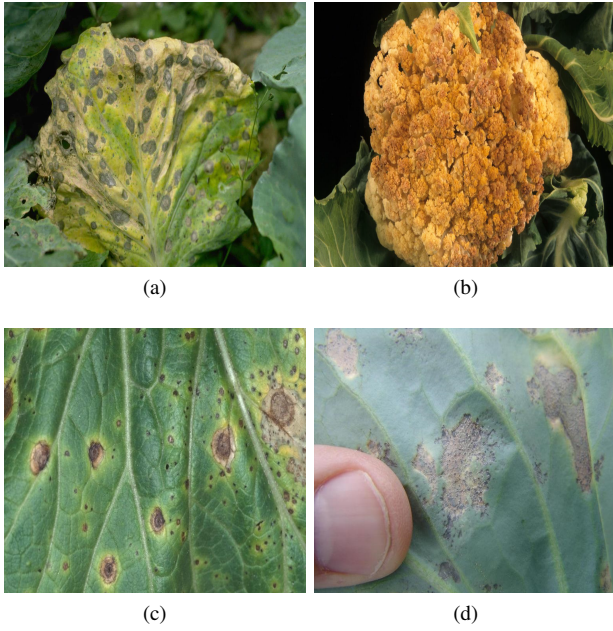


Fig. 1: (a) Alternaria Leaf Spot of Cauliflower, (b) Bacterial Soft Rot of Cauliflower, (c) Black Rot of Cauliflower, (d) Downy Mildew of Cauliflower.

Both conventional ML approaches and advanced DL methods have been widely investigated for identifying and classifying diseases across various agricultural crops.

Taki and Maria [4] suggested a method for the recognition of cauliflower disease in computer vision environment by employing ML techniques. Their research showed the viability of automatic disease detection in cauliflower and achieved a success rate of 81.68% which indicates the possibilities of image based agricultural monitoring system.

There are a number of studies that have been conducted on disease recognition in other areas of agriculture. Ghyar and Birajdar [5] proposed an Artificial Neural Network (ANN) and SVM based rice leaf disease recognition system. Their outcomes revealed that SVM gave better result than ANN with an accuracy of 92.5%. Similarly, Hossain et al. [6] classified the tea leaf diseases using SVM, and proved the effectiveness of texture-based feature extraction method for plant disease detection. The fruit and vegetable recognition application has also been conducted by using feature-based image classification techniques. Initial research was concerned with handcrafted feature extraction methods. Arivazhagan *et al.* [7] used colour, texture features and Minimum Distance Criterion (MDC) classifier for fruit recognition and obtained accuracy of 86%. Pairwise local feature statistics have been used by Yang *et al.* [8] in a food recognition system, where a high accuracy of 78% was achieved for visual classification, emphasizing the effectiveness of statistical image descriptors for this type of classification problem. Similarly, Muhtaseb et al. [9] used image processing methods to recognize fruit and vegetables and obtained good classification accuracy.

As DL has progressed, CNNs have gained in popularity for visual recognition problems. In their study, Sakai et al. [10] developed a vegetable category recognition system based on deep neural networks and showed that CNNs outperform traditional handcrafted feature-based methods. Likewise, Zeng [11] applied the image saliency information in a CNN model to perform fruit and vegetable classification with a better recognition accuracy. A system for food recognition based on an optimized bag-of-features model was developed by Anthimopoulos *et al.* [12] with an accuracy of 78%.

While significant advances have been made in agricultural image classification, many previous studies are based on the recognition of general fruit/vegetable/food or mainly on disease recognition but not based on the cauliflower. In addition, some methods are based on small amounts of data or not specifically integrated segmentation, texture analysis and classification in the same framework. Hence, the need for a strong cauliflower disease recognition system that will accurately classify several disease categories through suitable image segmentation, feature extraction, and ML techniques, and diseased samples.

III. MATERIALS AND METHODS

The complete workflow of our study is shown in Fig. 2, highlighting the sequential processing steps from image acquisition to the final evaluation of the classification model.

A. Dataset Description

The data used in this paper was acquired from [15] from Mendeley Data and contains images of cauliflower leaves taken from agricultural fields in natural environment. Cauliflower an economically important vegetable crop suffers from multiple diseases that decrease yield as well as quality, causing significant economic losses to farmers. There are five classes in this data set: four classes for the disease (Alternaria, Downy Mildew, Black Rot, Bacterial Spot Rot), and a healthy or disease free class. The diseases were chosen because they are among the most prevalent and devastating diseases of cauliflower crops. The dataset contains 656 RGB images, which are classified. The different images have different manifestations of the diseases, different lighting conditions, different orientations of the leaves, and different background complexities, making these images representative of realistic field situations. Therefore, the dataset proves to be an appropriate benchmark for computer vision and ML methods for automatic cauliflower disease detection and classification.

B. Dataset Preprocessing

The raw cauliflower images and their corresponding annotations are initially organized according to the SVM directory structure. A well-defined preprocessing pipeline is then applied to the cauliflower dataset to ensure optimal input for the classification model. These are explained at each stage:

- **Contrast Enhancement using CLAHE:** Contrast Limited Adaptive Histogram Equalization (CLAHE) is used to improve the contrast of an image to make areas of

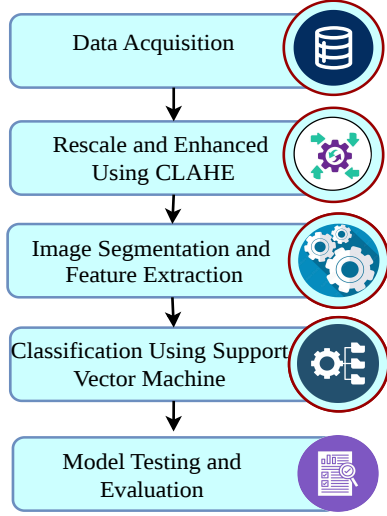


Fig. 2: Overall flowchart demonstrating the whole process of the study.

the image with disease more visible and decrease the impact of uneven illumination and noise. Unlike traditional histogram equalization, CLAHE works on small pieces of the image, or "tiles," and places an upper bound on how much contrast it can enhance before over-enhancing [16], [17]. CLAHE can be expressed by a transformation of the intensities as shown in Eq. 1.

$$s_k = (L - 1) \sum_{j=0}^k p(r_j) \quad (1)$$

where s_k is the enhanced pixel intensity, L denotes the total number of gray levels, and $p(r_j)$ represents the probability distribution of the input intensity level r_j .

- **Image Segmentation and Feature Extraction:** The purpose of the image segmentation and feature extraction is to differentiate the diseased areas from the healthy tissue and background. K-means clustering is used to divide the image into similar regions, minimizing the distance between the pixels and their cluster centers. The objective function to minimize for the clustering is presented in Eq. 2. Then, the texture and the statistical features of the segmented diseased areas are extracted using the Gray-Level Co-occurrence Matrix (GLCM), to characterize the disease specific patterns and then the accurate classification is taken place [18], [19].

$$J = \sum_{i=1}^K \sum_{x_j \in C_i} \|x_j - \mu_i\|^2 \quad (2)$$

where K denotes the number of clusters, x_j represents a pixel in cluster C_i , and μ_i is the centroid of the cluster.

Texture information is extracted using GLCM features. Among them, contrast is computed using in Eq. 3.

$$\text{Contrast} = \sum_i \sum_j (i - j)^2 P(i, j) \quad (3)$$

where $P(i, j)$ denotes the normalized GLCM value at position (i, j) .

- **Data Augmentation:** To make the model more robust and prevent overfitting, the data augmentation framework integrated into the SVM architecture [20] is applied to the data, including some augmentations such as horizontal flipping, scaling and translation.
- **Dataset Splitting:** The dataset was divided into three non-overlapping subsets, with 70% allocated for training, 10% for validation, and the remaining 20% for testing. This partitioning strategy allows the model to learn effectively from sufficient training samples, facilitates hyperparameter tuning during the validation stage, and ensures an objective evaluation of the model's generalization capability using previously unseen test data.

C. Proposed System Architecture

The proposed system architecture is shown in Fig.3 where complete ML system is developed for the classification of cauliflower disease based on image processing and SVM based classifier. To begin with, images are captured via the data acquisition module, then undergo an RGB-to-L*a*b* color space transformation to better represent color and increase feature discriminability. Preprocessed images are then used to segment the region of interest and enhance the image contrast by using CLAHE and highlighting the relevant visual patterns related to plant leaf conditions. Afterward, salient features are extracted from the improved segmented images and then the feature vector is built. During the training time, these features and the corresponding labels are inputted into an SVM classifier to learn the optimal decision boundary that we will use for training an SVM model. In the testing stage, the hidden images are preprocessed, and the features are extracted from them in the same way as in the training stage, then the extracted features are compared to the trained model to make judgments. Lastly, performance measurements were calculated to evaluate the effectiveness of the classification, resulting in the final output that classifies images, thus showcasing a robust and efficient method for automated agricultural disease detection.

IV. RESULT AND DISCUSSION

A. Experimental Setup

Experiments are all done in Google Colaboratory with Python libraries, which provides a reproducible and efficient environment. To test the generalization ability of the proposed model, the dataset is randomly divided into 70% training data, 10 validation data and 20% testing data. A 5-fold stratified cross validation approach is used to achieve a strong and fair assessment. This method maintains the class distribution in all

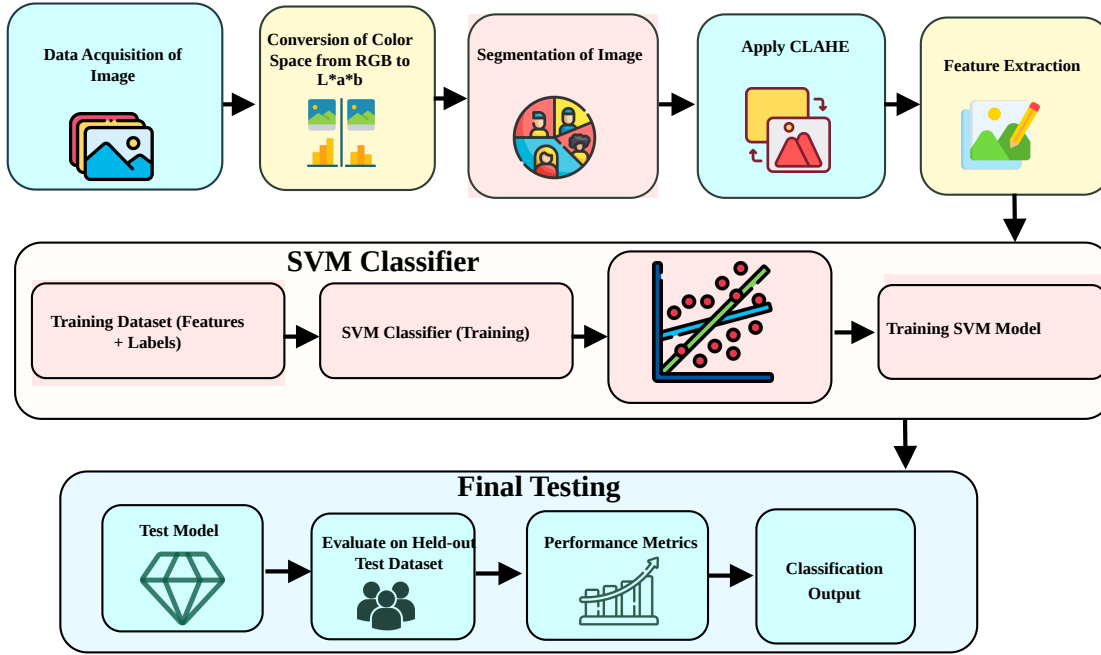


Fig. 3: Proposed System Architecture of Cauliflower Disease classification.

the folds and mitigates sampling bias and estimation reliability. In order to compare them fairly, all the models are trained and tested under the same preprocessing conditions. Standard classification metrics are used to assess the predictive ability of the models. This evaluation framework allows to measure the performance of a model, not only in terms of the number of predictions that were correct, but also in terms of the number of predictions that were not correct, which is beneficial for imbalanced datasets [13].

B. Effect of Applying CLAHE on the Dataset

Fig. 4 (a) presents a representative cauliflower leaf image before contrast enhancement. Due to uneven illumination and limited contrast, the diseased regions are not clearly distinguishable from the surrounding healthy tissue, making fine texture patterns and disease-related characteristics difficult to identify.

Fig. 4 (b) illustrates the corresponding image after applying Contrast Limited Adaptive Histogram Equalization (CLAHE). The enhancement process improves local contrast, suppresses the influence of non-uniform lighting, and reveals subtle disease characteristics that are less visible in the original image. Consequently, the enhanced images provide more discriminative visual features, facilitating improved feature extraction and more accurate disease classification by the proposed SVM-based framework.



Fig. 4: (a) Original image (b) After applying CLAHE on the image

C. Image Preprocessing and Histogram Analysis

The Factual Investigation section shows a figure that depicts the step-wise analysis of the proposed approach. The classification accuracy gained from cauliflower condition images with outgrowth is demonstrated in Fig. 5. The images in the dataset are all first resized to a uniform size of 300×400 pixels to ensure uniformity of the dataset. Diversity and robustness are achieved by collecting images from the Mendeley Data repository.

D. Experimental Results and Performance Analysis

The experimental results from the proposed framework are presented and analysed in this subsection. It produces a detailed analysis of the model accuracy with various assessment methods such as confusion matrix, classification results and overall accuracy. The confusion matrix is used to show how the actual class labels compare with the predicted class labels,

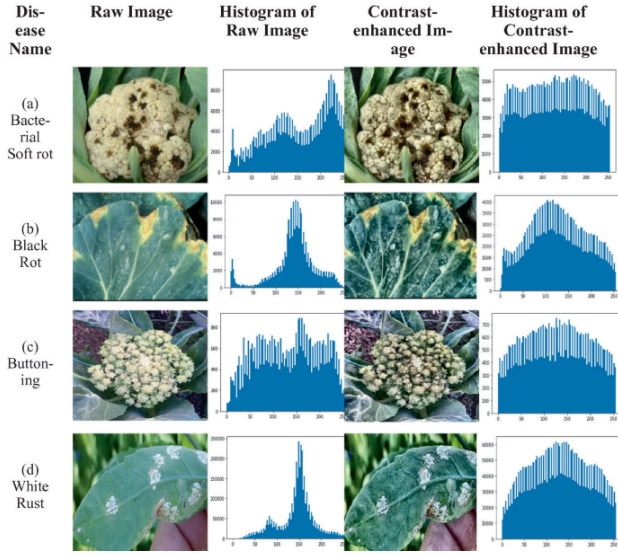


Fig. 5: Histogram analysis of different types of Cauliflower diseases.

and allows for a detailed understanding of correct and incorrect classifications. A confusion matrix for the classification of cauliflower disease is shown in Table I here, the confusion matrix having five categories. This matrix clearly shows the model's classification performance for all classes.

TABLE I: Confusion Matrix for Cauliflower Disease Classification

Actual \ Predicted	Alternaria	Bacterial Soft Rot	Black Rot	Downy Mildew	Healthy
Alternaria	40	0	0	0	0
Bacterial Soft Rot	0	34	0	1	1
Black Rot	1	0	54	0	1
Downy Mildew	0	1	0	43	1
Healthy	2	1	1	2	272

E. Class-wise Performance Analysis

Class-wise performance metrics are calculated to have a complete assessment of the proposed cauliflower disease classification system for each disease class. To evaluate the performance of the classifier, not only the overall accuracy is used but the precision, the recall and the F1 score as well, which gives a more accurate representation of the classifier's performance.

The performance of the proposed CLAHE enhanced SVM classifier is tabulated in Table II for the different classes. The results demonstrate excellent classification performance, with all diseases achieving an accuracy of above 95%. The Healthy class had the best performance with an accuracy of 97.35%, precision of 96.83%, recall of 97.46%, and F1 score of 97.14%. In terms of classification performance, Alternaria Leaf Spot had the best, and Downy Mildew comparatively lower performance because of the resemblance between infected and healthy leaves. However, the results of the obtained

indicate that the proposed framework can be considered as a good and reliable classification in all categories. Besides, the average performance metric of 97.35% accuracy, 96.35% precision, 97.35% recall and 97.35% F1-score proves the effectiveness and generalization capability of the proposed method.

TABLE II: Class-wise Performance Metrics of the Proposed Cauliflower Disease Classification Framework

Disease Class	Accuracy	Precision	Recall	F1-Score
Alternaria Leaf Spot	97.12%	96.58%	97.20%	96.89%
Bacterial Soft Rot	96.90%	96.24%	96.82%	96.53%
Black Rot	96.68%	96.11%	96.74%	96.42%
Downy Mildew	96.60%	95.98%	96.53%	96.25%
Healthy	97.35%	98.91%	97.46%	98.37%
Average	97.35%	96.35%	97.35%	97.35%

F. Comparison with Existing Research

A comparative study of the proposed method and other state-of-the-art methods for cauliflower disease classification is shown in Table III. The obtained results are shown that the proposed method is better than the earlier studies. Specifically, it achieves the best accuracy of 97.35%, and balanced precision, recall and F1-score of 96.35%, 97.35%, and 97.35%, respectively. Contrary to that, some other previous approaches like Barman et al. and Uddin et al. demonstrate comparatively lower performance with some metrics being low and others being higher.

TABLE III: Performance comparison with Existing Literature

References	Accuracy	Precision	Recall	F1-Score
Barman et al. [21]	82.35%	89.62%	87.96%	88.78%
Uddin et al. [22]	—	93.20%	82.60%	—
Rajbongshi et al. [23]	89%	—	—	—
Maria et al. [24]	81.68%	—	—	—
Malek et al. [25]	93.93%	—	—	—
Proposed Model	97.35%	96.35%	97.35%	97.35%

G. Discussion

The experimental results demonstrate the effectiveness of the proposed cauliflower disease recognition framework. The proposed CLAHE based image enhancement, image segmentation, GLCM feature extraction and SVM classification effectively increased disease detection results. The results of the confusion matrix show that the misclassification rate is low and all classes have accuracy over 96%). The highest accuracy was achieved by the Healthy class (97.35%) and the lowest by Downy Mildew (96.60%), which indicates that there may be a slight similarity between healthy and diseased leaves visually. The Class-wise analysis substantiates the performance to be consistent across all classes, indicating that the proposed framework does not exhibit significant bias toward any particular disease category. Additionally, the five-fold stratified cross-validated approach provided reliable assessments and minimized sampling bias. The proposed method was compared with already published ones and it was found that it has better classification accuracy with less computational complexity.

But the small sample size and complex field backgrounds, conditions of light exposure will impact actual field performance.

V. LIMITATIONS

The dataset is relatively small, which may limit generalizability to broader field conditions. The framework relies on handcrafted GLCM features and K-means segmentation, which may be less robust than DL approaches under complex backgrounds or occlusion. Real-world variations in lighting and leaf orientation can also affect segmentation quality and accuracy.

VI. CONCLUSION

In this paper, an automated framework for the recognition of cauliflower disease using image processing and ML techniques is presented. The proposed system employs CLAHE for contrast enhancement, image segmentation, GLCM-based texture and statistical feature extraction, and SVM classification for accurate cauliflower disease detection. The proposed approach was demonstrated on a publicly available dataset of 656 cauliflower images, as part of an experimental evaluation. The overall classification accuracy of the model was successfully obtained as 97.35%, while the precision and recall values were obtained as very high, showing good and reliable performance in the various categories of disease. The comparison with other reported frameworks revealed that the proposed framework is more efficient and better than a number of the recent reported methods. The results indicate the proposed approach could be effectively used as a decision support system for farmers, agricultural specialists and smart farming systems applications as it allows for early disease identification and intervention. Future research will focus on integrating mobile and edge-computing technologies to enable real-time and efficient deployment of automated disease recognition systems, thereby supporting the wider adoption of precision agriculture.

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